INCH-POUND
MIL-DTL-83503/21C
17 June 1998
SUPERSEDING
MIL-C-83503/21B
1 July 1983

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, FLAT CABLE, NONENVIRONMENTAL
HEADER, STRAIGHT THRU (.100" SPACING) WITH POLARIZING FEATURE
SOLDERLESS WRAPPOST OR PWB TERMINATION

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-DTL-83503.

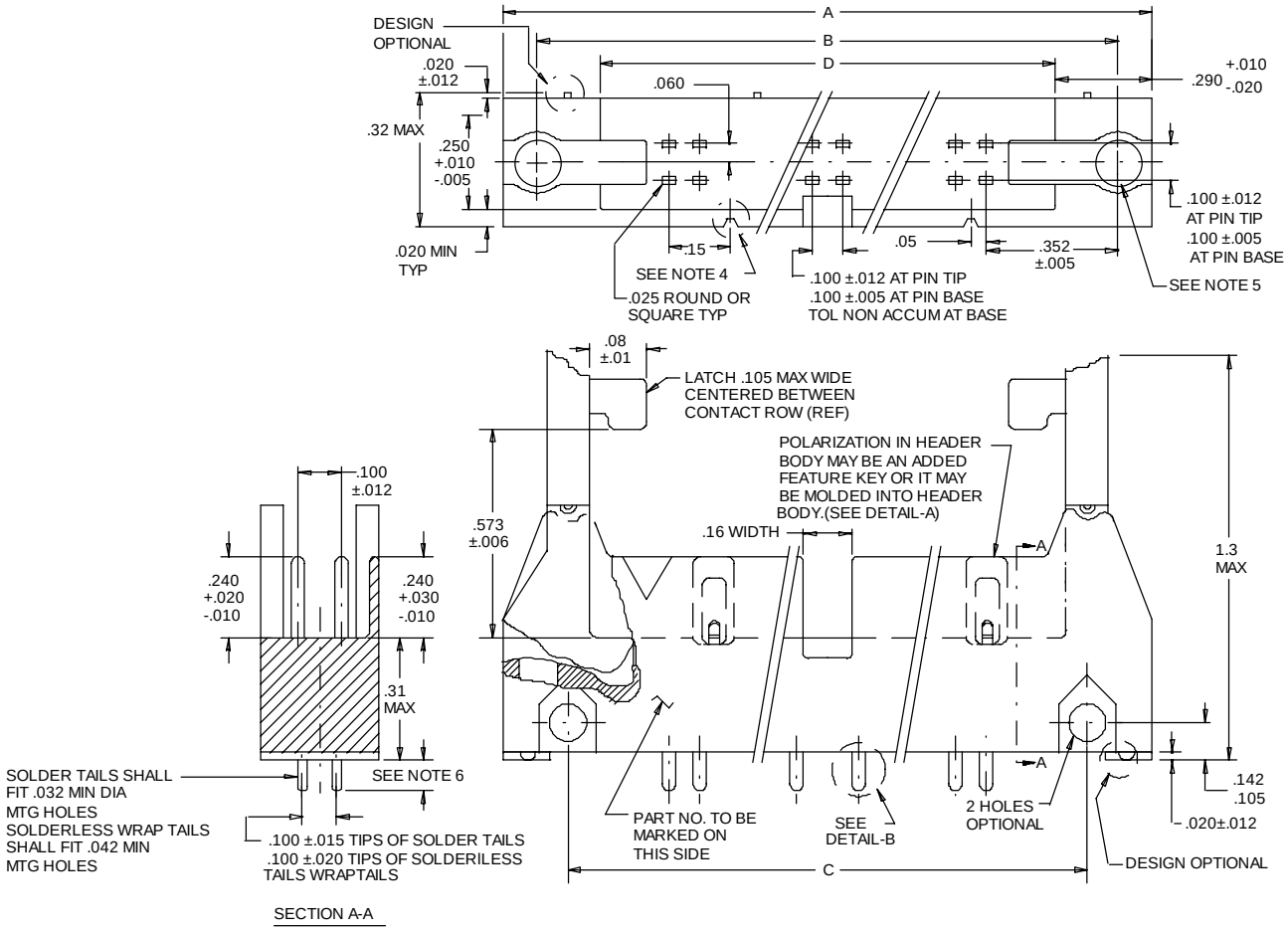

Figure 1. Dimensions and configuration.

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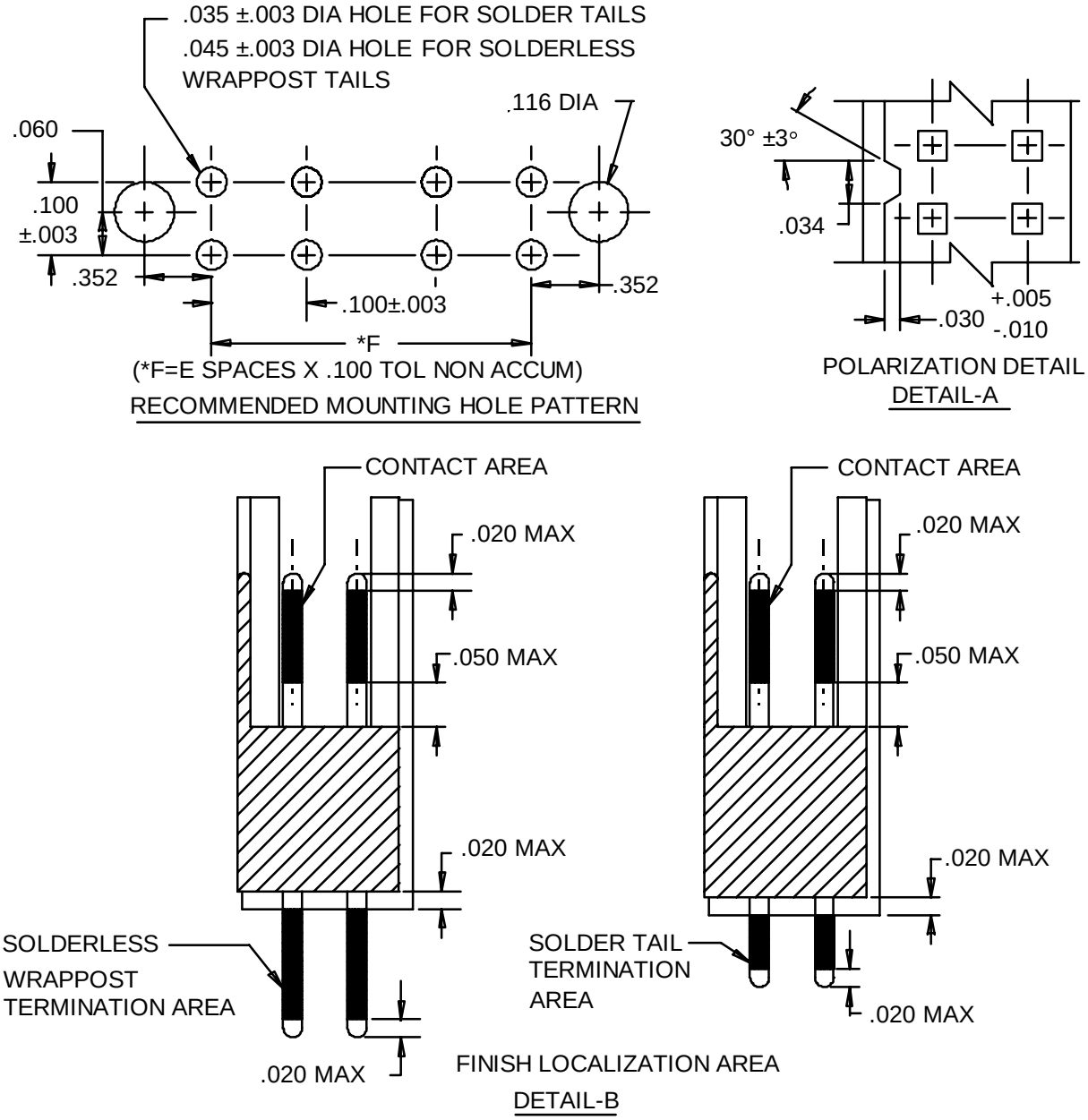

FIGURE 1. Dimensions and configuration - Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.003	0.08	.025	0.64	.089	2.26	.250	6.35
.005	0.13	.030	0.76	.100	2.54	.290	7.37
.006	0.15	.034	0.86	.105	2.67	.31	7.9
.010	0.25	.035	0.89	.116	2.95	.32	8.13
.01	0.3	.045	1.14	.125	3.18	.352	8.94
.012	0.30	.050	1.3	.15	3.8	.573	14.55
.015	0.38	.060	1.52	.152	3.86	.610	15.49
.020	0.51	.062	1.57	.178	4.52	1.3	33.0
		.08	2.0	.240	6.10		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25 mm.
3. Unless otherwise specified tolerances are ± .005 (0.13 mm) inch for three place decimals and ± .01 (0.3 mm) inch for two place decimals.
4. The polarizing feature on this end is non-existent on the 10 and 14 position connectors.
5. Two thread cutting screws, No. 4-24 (1/4 inch long) type BG are provided.
6. Pin length for printed wiring board (PWB) thickness of:

   .062 (1.57 mm) inch thick; pin length shall be .089 (2.26 mm) inch minimum and .116 (2.95 mm) inch maximum
   .125 (3.18 mm) inch thick; pin length shall be .152 (3.86 mm)minimum and .178 (4.52 mm) inch maximum.
   Solderless wrappost, pin length shall be .610 ±.015 (15.49 ±0.38 mm) inch.

FIGURE 1. Dimensions and configuration - Continued.

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TABLE I. Dash numbers and characteristics

Dash Number	No. of pins	Dimensions 1/ A (mm)	B (mm)	C (mm)	D (mm) +.030 (0.76) -.000	E (mm)	F (mm)	Termination, right angle contact type
-01	10	1.26 (32.0)	1.10 (27.9)	.86 (21.84)	.690 (17.53)	4	.400 (10.16)	.062 PWB
-02	10							.125 PWB
-03	10							Wrap post
-04	14	1.46 (37.1)	1.30 (33.0)	1.06 (26.92)	.890 (21.34)	6	.600 (15.24)	.062 PWB
-05	14							.125 PWB
-06	14							Wrap post
-07	16	1.56 (39.6)	1.40 (35.6)	1.16 (29.46)	.990 (25.15)	7	.700 (17.78)	.062 PWB
-08	16							.125 PWB
-09	16							Wrap post
-10	20	1.76 (44.7)	1.60 (40.6)	1.36 (34.54)	1.190 (30.23)	9	.900 (22.86)	.062 PWB
-11	20							.125 PWB
-12	20							Wrap post
-13	26	2.06 (52.3)	1.90 (48.3)	1.66 (42.16)	1.490 (37.85)	12	1.200 (30.48	.062 PWB
-14	26							.125 PWB
-15	26							Wrap post
-16	34	2.46 (62.5)	2.30 (58.4)	2.06 (52.32)	2.06 (52.32)	16	1.600 (40.64)	.062 PWB
-17	34							.125 PWB
-18	34							Wrap post
-19	40	2.76 (70.1	2.60 (66.0)	2.36 (59.94)	2.36 (59.94)	19	1.900 (48.26)	.062 PWB
-20	40							.125 PWB
-21	40							Wrap post
-22	50	3.26 (82.8)	3.10 (78.7)	2.86 (72.64)	2.86 (72.64)	24	2.400 (60.96)	.062 PWB
-23	50							.125 PWB
-24	50							Wrap post
-25	60	3.76 (95.5)	3.60 (91.4)	3.36 (85.34)	3.36 (85.34)	29	2.900 (73.66)	.062 PWB
-26	60							.125 PWB
-27	60							Wrap post
-28	64	3.96 (100.6)	3.80 (96.5)	3.56 (90.42)	3.390 (86.11)	31	3.100 (78.74)	.062 PWB
-29	64							.125 PWB
-30	64							Wrap post

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REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figure 1 and table I.

Temperature range: -55° to +120°C.

Voltage rating: 300 V rms.

Current rating: 1 ampere.

Materials:

Contacts: Phosphor bronze in accordance with ASTM-B-139, C51000, brass per CDA260, spring temper; or copper alloy CA-725 in accordance with ASTM-B-122.

Contact plating:
Contact area: Gold in accordance with MIL-DTL-83503.
Solderless wrap post terminal area: In accordance with MIL-STD-1130.
Solder tail termination area: Gold , or tin-lead in accordance with MIL-DTL-83503.

Housing: Glass field polyester in accordance with ASTM PS 2, type GPT-15F, GPT-20F, GPT-30F.

Accessories: Parts may be constructed of polyester per ASTM PS 2 or other materials specified in MIL-DTL-83503, provided these parts do not come in contact with current carrying components.

Polarization key: Shall be permanently attached to header (added feature key shall be bonded to header).

Mating connector : see MIL-DTL-83503/7.

Solderless wrap post contacts: Geometry parallelism and tip configuration shall be in accordance with MIL-STD-1130.

Solderless wrap post contact wire accommodation: 26 or 28 AWG solid.

Dielectric withstanding voltage:

(at sea level) : 500 V rms, minimum.

(at altitude): 200 V rms, minimum.

Cable retention: Not applicable.

Connector-cable flexing: Not applicable.

Contact resistance: 50 milliohms, maximum.

Latch retention: The latch shall retain the mating connector when the cable retention force is applied to the mating connector cable. The latch shall also function as an ejector to eject mating connector.

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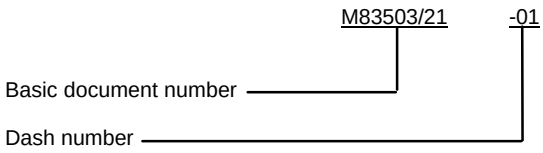
Marking:

Part or identifying number (PIN): The PIN shall consist of this specification sheet and the dash number from table I.

Example



Contact identification: Molded triangle "Δ " or notch to show number 1 contact, or raised numbers to show position number " 1".

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians:
Air Force - 85

Preparing activity:
DLA - CC

(Project 5935-4110-04)

Review activities:
Air Force - 17, 99

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